INCH-POUND
MIL-R-22684/4B
6 August 2001
SUPERSEDING
MIL-R-22684/4A
5 October 1972

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, INSULATED, STYLE RL42

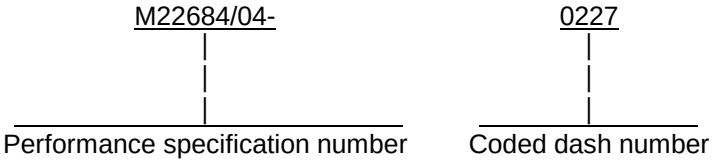
This specification is inactive for design after 5 October 1972. Use MIL-PRF-22684/8C.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for insulated, film, fixed resistors of 2-percent and 5-percent resistance tolerance. These resistors are capable of full-load operation at an ambient temperature of 70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/°C).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:



The coded dash number is derived in accordance with MIL-PRF-22684.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DSCC-VAT, P.O. Box 3990 , Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the basic document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-22684 - Resistors, Fixed, Film (Insulated), General Specification.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DoDSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related association specification, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-22684.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 2 watts based on full load operation at an ambient temperature of 70°C.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 500 volts.

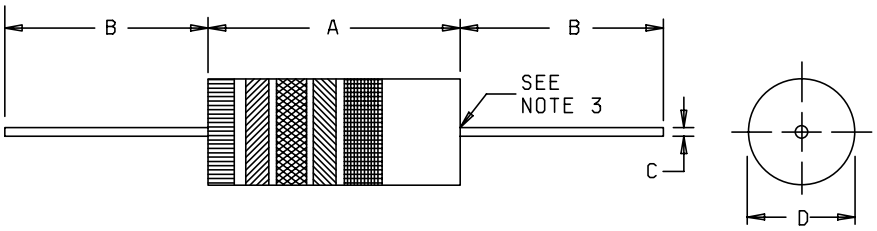
3.5 Resistance values and resistance tolerances. The minimum and maximum standard resistance values and associated resistance tolerances shall be as listed in table I.

3.6 Terminal type. The terminal type available shall be in accordance with MIL-PRF-22684 and table I.

3.7 Dielectric withstanding voltage. Resistors shall be tested as specified in MIL-PRF-22684. The magnitude of test voltage shall be as follows:

Atmospheric pressure - 1,000 volts rms.
Barometric pressure - 500 volts rms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.



Ltr	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	0.648	0.728	16.46	18.49
B	1.375	1.625	34.92	41.28
C	.040	0.050	1.02	1.27
D	0.280	0.336	7.11	8.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

3.8 Insulation resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the insulation resistance shall be not less than 100 megohms.

3.9 Moisture resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 2.0 percent.

3.10 Life. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 3.0 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Table I. PIN designation.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL42S100G	10	G	S	0001	RL42S121G	120	G	S	0053
RL42S100J	10	J	S	0002	RL42S121J	120	J	S	0054
RL42S110G	11	G	S	0003	RL42S131G	130	G	S	0055
RL42S110J	11	J	S	0004	RL42S131J	130	J	S	0056
RL42S120G	12	G	S	0005	RL42S151G	150	G	S	0057
RL42S120J	12	J	S	0006	RL42S151J	150	J	S	0058
RL42S130G	13	G	S	0007	RL42S161G	160	G	S	0059
RL42S130J	13	J	S	0008	RL42S161J	160	J	S	0060
RL42S150G	15	G	S	0009	RL42S181G	180	G	S	0061
RL42S150J	15	J	S	0010	RL42S181J	180	J	S	0062
RL42S160G	16	G	S	0011	RL42S201G	200	G	S	0063
RL42S160J	16	J	S	0012	RL42S201J	200	J	S	0064
RL42S180G	18	G	S	0013	RL42S221G	220	G	S	0065
RL42S180J	18	J	S	0014	RL42S221J	220	J	S	0066
RL42S200G	20	G	S	0015	RL42S241G	240	G	S	0067
RL42S200J	20	J	S	0016	RL42S241J	240	J	S	0068
RL42S220G	22	G	S	0017	RL42S271G	270	G	S	0069
RL42S220J	22	J	S	0018	RL42S271J	270	J	S	0070
RL42S240G	24	G	S	0019	RL42S301G	300	G	S	0071
RL42S240J	24	J	S	0020	RL42S301J	300	J	S	0072
RL42S270G	27	G	S	0021	RL42S331G	330	G	S	0073
RL42S270J	27	J	S	0022	RL42S331J	330	J	S	0074
RL42S300G	30	G	S	0023	RL42S361G	360	G	S	0075
RL42S300J	30	J	S	0024	RL42S361J	360	J	S	0076
RL42S330G	33	G	S	0025	RL42S391G	390	G	S	0077
RL42S330J	33	J	S	0026	RL42S391J	390	J	S	0078
RL42S360G	36	G	S	0027	RL42S431G	430	G	S	0079
RL42S360J	36	J	S	0028	RL42S431J	430	J	S	0080
RL42S390G	39	G	S	0029	RL42S471G	470	G	S	0081
RL42S390J	39	J	S	0030	RL42S471J	470	J	S	0082
RL42S430G	43	G	S	0031	RL42S511G	510	G	S	0083
RL42S430J	43	J	S	0032	RL42S511J	510	J	S	0084
RL42S470G	47	G	S	0033	RL42S561G	560	G	S	0085
RL42S470J	47	J	S	0034	RL42S561J	560	J	S	0086
RL42S510G	51	G	S	0035	RL42S621G	620	G	S	0087
RL42S510J	51	J	S	0036	RL42S621J	620	J	S	0088
RL42S560G	56	G	S	0037	RL42S681G	680	G	S	0089
RL42S560J	56	J	S	0038	RL42S681J	680	J	S	0090
RL42S620G	62	G	S	0039	RL42S751G	750	G	S	0091
RL42S620J	62	J	S	0040	RL42S751J	750	J	S	0092
RL42S680G	68	G	S	0041	RL42S821G	820	G	S	0093
RL42S680J	68	J	S	0042	RL42S821J	820	J	S	0094
RL42S750G	75	G	S	0043	RL42S911G	910	G	S	0095
RL42S750J	75	J	S	0044	RL42S911J	910	J	S	0096
RL42S820G	82	G	S	0045	RL42S102G	1,000	G	S	0097
RL42S820J	82	J	S	0046	RL42S102J	1,000	J	S	0098
RL42S910G	91	G	S	0047	RL42S112G	1,100	G	S	0099
RL42S910J	91	J	S	0048	RL42S112J	1,100	J	S	0100
RL42S101G	100	G	S	0049	RL42S122G	1,200	G	S	0101
RL42S101J	100	J	S	0050	RL42S122J	1,200	J	S	0102
RL42S111G	110	G	S	0051	RL42S132G	1,300	G	S	0103
RL42S111J	110	J	S	0052	RL42S132J	1,300	J	S	0104

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Table I. PIN designation - continued.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			0159
RL42S152G	1,500	G	S	0105	RL42S203G	20,000	G	S	0160
RL42S152J	1,500	J	S	0106	RL42S203J	20,000	J	S	0161
RL42S162G	1,600	G	S	0107	RL42S223G	22,000	G	S	0162
RL42S162J	1,600	J	S	0108	RL42S223J	22,000	J	S	
RL42S182G	1,800	G	S	0109	RL42S243G	24,000	G	S	0163
RL42S182J	1,800	J	S	0110	RL42S243J	24,000	J	S	0164
RL42S202G	2,000	G	S	0111	RL42S273G	27,000	G	S	0165
RL42S202J	2,000	J	S	0112	RL42S273J	27,000	J	S	0166
RL42S222G	2,200	G	S	0113	RL42S303G	30,000	G	S	0167
RL42S222J	2,200	J	S	0114	RL42S303J	30,000	J	S	0168
RL42S242G	2,400	G	S	0115	RL42S333G	33,000	G	S	0169
RL42S242J	2,400	J	S	0116	RL42S333J	33,000	J	S	0170
RL42S272G	2,700	G	S	0117	RL42S363G	36,000	G	S	0171
RL42S272J	2,700	J	S	0118	RL42S363J	36,000	J	S	0172
RL42S302G	3,000	G	S	0119	RL42S393G	39,000	G	S	0173
RL42S302J	3,000	J	S	0120	RL42S393J	39,000	J	S	0174
RL42S332G	3,300	G	S	0121	RL42S433G	43,000	G	S	0175
RL42S332J	3,300	J	S	0122	RL42S433J	43,000	J	S	0176
RL42S362G	3,600	G	S	0123	RL42S473G	47,000	G	S	0177
RL42S362J	3,600	J	S	0124	RL42S473J	47,000	J	S	0178
RL42S392G	3,900	G	S	0125	RL42S513G	51,000	G	S	0179
RL42S392J	3,900	J	S	0126	RL42S513J	51,000	J	S	0180
RL42S432G	4,300	G	S	0127	RL42S563G	56,000	G	S	0181
RL42S432J	4,300	J	S	0128	RL42S563J	56,000	J	S	0182
RL42S472G	4,700	G	S	0129	RL42S623G	62,000	G	S	0183
RL42S472J	4,700	J	S	0130	RL42S623J	62,000	J	S	0184
RL42S512G	5,100	G	S	0131	RL42S683G	68,000	G	S	0185
RL42S512J	5,100	J	S	0132	RL42S683J	68,000	J	S	0186
RL42S562G	5,600	G	S	0133	RL42S753G	75,000	G	S	0187
RL42S562J	5,600	J	S	0134	RL42S753J	75,000	J	S	0188
RL42S622G	6,200	G	S	0135	RL42S823G	82,000	G	S	0189
RL42S622J	6,200	J	S	0136	RL42S823J	82,000	J	S	0190
RL42S682G	6,800	G	S	0137	RL42S913G	91,000	G	S	0191
RL42S682J	6,800	J	S	0138	RL42S913J	91,000	J	S	0192
RL42S752G	7,500	G	S	0139					
RL42S752J	7,500	J	S	0140		Megohm			
RL42S822G	8,200	G	S	0141	RL42S104G	0.10	G	S	0193
RL42S822J	8,200	J	S	0142	RL42S104J	0.10	J	S	0194
RL42S912G	9,100	G	S	0143	RL42S114G	0.11	G	S	0195
RL42S912J	9,100	J	S	0144	RL42S114J	0.11	J	S	0196
RL42S103G	10,000	G	S	0145	RL42S124G	0.12	G	S	0197
RL42S103J	10,000	J	S	0146	RL42S124J	0.12	J	S	0198
RL42S113G	11,000	G	S	0147	RL42S134G	0.13	G	S	0199
RL42S113J	11,000	J	S	0148	RL42S134J	0.13	J	S	0200
RL42S123G	12,000	G	S	0149	RL42S154G	0.15	G	S	0201
RL42S123J	12,000	J	S	0150	RL42S154J	0.15	J	S	0202
RL42S133G	13,000	G	S	0151	RL42S164G	0.16	G	S	0203
RL42S133J	13,000	J	S	0152	RL42S164J	0.16	J	S	0204
RL42S153G	15,000	G	S	0153	RL42S184G	0.18	G	S	0205
RL42S153J	15,000	J	S	0154	RL42S184J	0.18	J	S	0206
RL42S163G	16,000	G	S	0155	RL42S204G	0.20	G	S	0207
RL42S163J	16,000	J	S	0156	RL42S204J	0.20	J	S	0208
RL42S183G	18,000	G	S	0157	RL42S224G	0.22	G	S	0209
RL42S183J	18,000	J	S	0158	RL42S224J	0.22	J	S	0210

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

Table I. PIN designation - continued.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Megohm					Megohm			
RL42S244G	0.24	G	S	0211	RL42S624G	0.62	G	S	0231
RL42S244J	0.24	J	S	0212	RL42S624J	0.62	J	S	0232
RL42S274G	0.27	G	S	0213	RL42S684G	0.68	G	S	0233
RL42S274J	0.27	J	S	0214	RL42S684J	0.68	J	S	0234
RL42S304G	0.30	G	S	0215	RL42S754G	0.75	G	S	0235
RL42S304J	0.30	J	S	0216	RL42S754J	0.75	J	S	0236
RL42S334G	0.33	G	S	0217	RL42S824G	0.82	G	S	0237
RL42S334J	0.33	J	S	0218	RL42S824J	0.82	J	S	0238
RL42S364G	0.36	G	S	0219	RL42S914G	0.91	G	S	0239
RL42S364J	0.36	J	S	0220	RL42S914J	0.91	J	S	0240
RL42S394G	0.39	G	S	0221	RL42S105G	1.0	G	S	0241
RL42S394J	0.39	J	S	0222	RL42S105J	1.0	J	S	0242
RL42S434G	0.43	G	S	0223	RL42S115G	1.1	G	S	0243
RL42S434J	0.43	J	S	0224	RL42S115J	1.1	J	S	0244
RL42S474G	0.47	G	S	0225	RL42S125G	1.2	G	S	0245
RL42S474J	0.47	J	S	0226	RL42S125J	1.2	J	S	0246
RL42S514G	0.51	G	S	0227	RL42S135G	1.3	G	S	0247
RL42S514J	0.51	J	S	0228	RL42S135J	1.3	J	S	0248
RL42S564G	0.56	G	S	0229	RL42S155G	1.5	G	S	0249
RL42S564J	0.56	J	S	0230	RL42S155J	1.5	J	S	0250-

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with Group A inspection and Group B inspection of MIL-PRF-22684 and as specified herein.

5. PACKAGING

5.1 Packaging. For acquisition purposes the packaging requirements shall be as specified in the contract or purchase order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes The notes specified in MIL-PRF-22684 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

c. Packaging requirements.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, AT, AV
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5905-1629-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete block1s 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-R-22684/4B**	**2. DOCUMENT DATE** (YYMMDD) 2001 August 6

3. DOCUMENT TITLE
RESISTORS, FIXED, FILM, INSULATED, STYLE RL42

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus ATTN: DSCC – VAT	b. TELEPHONE (Include Area Code) (1) Commercial (614) 692-0553 (2) DSN 850-0553
c. ADDRESS (Include Zip Code) P.O. 3990 Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:24Z
Check the source to verify that this is the current version before use.